

CS 550 Operating Systems, Spring 2026
Homework Assignment 5 (HW5)

Out: 3/15/2026 Sun.

Due: 3/22/2026 Sun. 23:59:59

Full points: 100

Note:

- There are 5 questions, each on a separate page.
- Read the submission instructions on the last page carefully and prepare the submission accordingly.

1. (20 points) Choose whether each of the following statements is true or false.

- (1) In C, memory allocated for a local variable inside a function is typically placed on the stack, and that memory is automatically reclaimed when the function returns.

(A) True (B) False

- (2) In C, `sizeof(ptr)` returns the size of the heap region pointed to by `ptr`.

(A) True (B) False

- (3) When allocating space for a copied C string, we need to allocate `strlen(src)+1` bytes via `malloc(strlen(src)+1)`.

(A) True (B) False

- (4) One major goal of memory virtualization is to give each process the illusion of its own private address space while still allowing the OS to enforce protection.

(A) True (B) False

- (5) In a system that uses dynamic relocation, the hardware computes a physical address by adding the base register value to the virtual address.

(A) True (B) False

- (6) In a system that uses dynamic relocation, if a virtual address is greater than or equal to the bounds value, the hardware should treat it as illegal.

(A) True (B) False

- (7) In a base-and-bounds system, the bounds register stores the largest legal physical address that the process may access.

(A) True (B) False

- (8) In dynamic relocation, increasing the base register by 100 increases the largest legal virtual address by 100.

(A) True (B) False

- (9) In a positively growing segment, a memory reference is legal only if the offset is strictly less than the bounds of that segment.

(A) True (B) False

- (10) If two segments have disjoint virtual address ranges, then they must also occupy adjacent physical memory ranges after translation.

(A) True (B) False

2. (20 points) A machine uses dynamic relocation (base-and-bounds). The virtual address space size is 512 bytes. The base register contains 4000, and the bounds register contains 120. For each of the following virtual addresses, state whether the access is legal. If legal, also give the translated physical address: 0, 50, 119, 120, 511.

$$(0) : 0 < 120 \quad \therefore \text{legal.}$$

$$PA : 4000 + 0 = 4000.$$

$$(50) : 50 < 120 \quad \therefore \text{legal}$$

$$PA = 4000 + 50 = 4050.$$

$$(119) : 119 < 120 \quad \therefore \text{legal}$$

$$PA = 4000 + 119 = 4119$$

$$(120) : 120 = 120 \quad \therefore \text{illegal}$$

$$(511) : 511 > 120 \quad \therefore \text{illegal}$$

3. (20 points) A machine uses dynamic relocation. For one process, virtual address 20 translates to physical address 3020, and virtual address 99 translates to physical address 3099.

(1) What is the value of the base register?

(2) What is the smallest possible value of the bounds register?

$$\begin{aligned} (1) \quad PA &= \text{base} + VA \Rightarrow \text{base} = PA - VA \\ &\Rightarrow \quad \quad \quad = 3020 - 20 \\ &\quad \quad \quad = 3000 \\ \text{base} &= 3000 \end{aligned}$$

(2) $\because$ 3099 is legal
 $\therefore$ the smallest bounds should be greater than 99
 $\therefore$ smallest bound is 100.

4. (20 points) A simple segmentation system supports two segments in a 128-byte virtual address space:

- Segment 0 (code+heap, grows positive): base = 1000, limit = 30
- Segment 1 (stack, grows negative): base = 2000, limit = 20

The top bit of the 7-bit virtual address selects the segment. The remaining 6 bits are the offset field.

(1) For each virtual address below, state whether it is legal: 0, 29, 30, 127, 110, 100.

(2) Translate the following legal virtual addresses into physical addresses: 5, 20, 127, 120, 108.

[1] Seg 0 base = 0 Seg 1 base = 64.
∵ 0, 29, 30 < 63 ∴ they are in Seg 0.
127, 110, 100 ≥ 64 ∴ they are in Seg 1
(0) = 0 - 0 = 0 < 30 ∴ legal (29) = 29 - 0 = 29 < 30 ∴ legal
(30) = 30 - 0 = 30 = 30 ∴ illegal (64) 127 - 127 = 0 < 20 ∴ legal
(110) 127 - 110 = 17 < 20 ∴ legal (120) 127 - 100 = 27 > 20 ∴ illegal

[2] 5, 20 < 63 segment 0.

127, 120, 108 > 64 segment 1

(5): 5 - 0 = 5

PA-5: 1000 + 5 = 1005

(20) 20 - 0 = 20

PA-20: 1000 + 20 = 1020

(127) 127 - 127 = 0

PA-127 = 2000 + 0 = 2000

(120) 127 - 120 = 7

PA-120 = 2000 + 7 = 2007

(108) 127 - 108 = 19

PA-108 = 2000 + 19 = 2019

5. (20 points) A machine uses two segments in a 128-byte virtual address space. The top bit selects the segment. Segment 0 grows positive, and Segment 1 grows negative. You are told:

VA 127 $\rightarrow$ PA 799

VA 120 $\rightarrow$ PA 792

VA 119 is illegal

Answer the following questions:

- (1) How many bits does a virtual address (VA) have?
- (2) What is the maximum size of a segment?
- (3) How many bits does the offset part of a VA have?
- (4) What are the base and limit values of Segment 1?

(1) $128 = 2^7$ $\therefore$ 7 bits.

(2) $128/2 = 64$ $\therefore$ max size = 64.

(3) top bit selects seg $\therefore$ offset = $7 - 1 = 6$ bit.

(4) $127 - 64 = 63$
 $120 - 64 = 56$

$799 - 63 = 736$
check: $736 + 56 = 792$
 $\therefore$ Base = 736.

$127 - 119 = 8$ $\therefore$ limit = 8

Submission instructions

1. Type your answers using the text editor you like. Remember to include the index number of each question.
2. Export the file to PDF format.
3. Name the PDF file based on your BU email ID. For example, if your BU email is “abc@binghamton.edu”, then the PDF file should be named as “hw5_abc.pdf”.
4. Failure to follow the above instructions will lead to 5 points off.
5. Submit the PDF file to Brightspace before the deadline.